

Exercise #7:

Blood regulation competition

April 17, 2026

This is a **mandatory** exercise that you need to pass to access the final exam in summer.

The Matlab file `competition.m` computes the red blood cell (RBC) dynamics in an idealized hexagonal capillary network. The average hematocrit of the whole network is kept constant ($\hat{H}_T = 0.25$) during the simulation, i.e. every RBC which leaves the network at the bottom-right will be immediately replaced by a new incoming RBC at the top-left. When RBCs reach a bifurcation, they will enter the capillary with the larger bulk flow velocity according to Yen et al. [1]. We further account for the Fåhræus effect by following formulae from Pries et al. [2].

Before starting the exercise:

- Run the simulation without making any changes (for this time with images).
- Observe the flow and hematocrit distribution that is plotted at the end of the simulation.
- What do you notice? Is the final distribution of RBC homogeneous?

Sixteenth ETH competition in blood flow regulation

The original network configuration is homogeneous, i.e., all vessels have equal length and radius. In this homogeneous configuration, the hematocrit distribution becomes inhomogeneous (after a transient phase). The goal of this assignment is to modify the network (by changing the vessel radii) such that the hematocrit distribution becomes more homogeneous. We will quantify the homogeneity of the hematocrit distribution by counting all the vessels that feature a mean hematocrit of $\overline{H}_T = 0.25 \pm 0.1$ (averaged from 10s to 20s of the simulation, *bin count*) and calculate the percentage to the total number of vessels. The goal is to increase this percentage from the 12,4% where no changes are applied.

To receive a passing grade you need to achieve at least 16% of vessels with $\overline{H}_T = 0.25 \pm 0.1$. The one who finds the **best** solution will also be **awarded** by the celestial commission for blood flow regulation. These are the rules:

- You can only change the vessel radii in the top left half of the capillary network (see Figure 1).
- You can only dilate or constrict the diameters in these capillaries by 40%. The number of changes however is not limited.
- The results must be repeatable.

The submission process is done over Moodle under the Quizzes section. You need to upload the following **six** files with your name **zipped** in one folder called **Figures**:

- `name_surname_Setting.mat`
- `name_surname_Results.mat`
- `name_surname_Standard_Deviation.png`
- `name_surname_Flow.png`

- name_surname_Hematocrit.png
- name_surname_Bin_Count.png

Those files are automatically saved in the Figure folder where you have your `competition.m` script. The easiest way to name them correctly is to run the routine `competition.m`, insert 'name_surname' in the test name input dialogue. The deadline is by May 22, 2026 at 23:59.

The Matlab file `competition.m` calls the following Matlab functions and data:

```
create\_network.m,
select\_capillaries.m,
compute\_network.m,
randcon.mat,
randleng.m,
bifurcation\_time.m,
plot\_network.m,
plot\_time\_averaged\_flow.m,
compute\_relative\_effective\_viscosity.m,
compute\_discharge\_from\_tube.m.
```

Within those functions you will find three matrices: The vertices p (Table 1), the capillaries c between those vertices (Table 2), and the red blood cells r (Table 3).

p	1	2	3	4	5	6	7
	x	y	P	nc	f	$H_{T,avg}$	f_{avg}
	coordinate	coordinate	pressure	number of connections to this point	flow rate across the node	time-averaged hematocrit	time-averaged flow rate

Table 1: Definition of the vertices p which set up the hexagonal network.

c	1	2	3	4	5	6	7	8	9	10
	p0	p1	T	n_{RBC}	u_{bulk}	L	H_T	a	H_D	vf
	start node	end node	transmissibility	number of RBCs inside this vessel	bulk flow velocity	vessel length	tube hematocrit	vessel radius	discharge hematocrit	velocity correction factor

Table 2: Definition of the capillaries c between two vertices p0 and p1.

r	1	2	3	4	5	6
	x	y	ic	s	T_{age}	T_{remain}
	coordinate	coordinate	capillary ID	position along the vessel (s=0: p0, s= $1.5 \cdot 10^{-5}$: p1)	age of the RBC	time until RBC reaches next bifurcation

Table 3: Definition of the red blood cells r which flow through the capillaries.

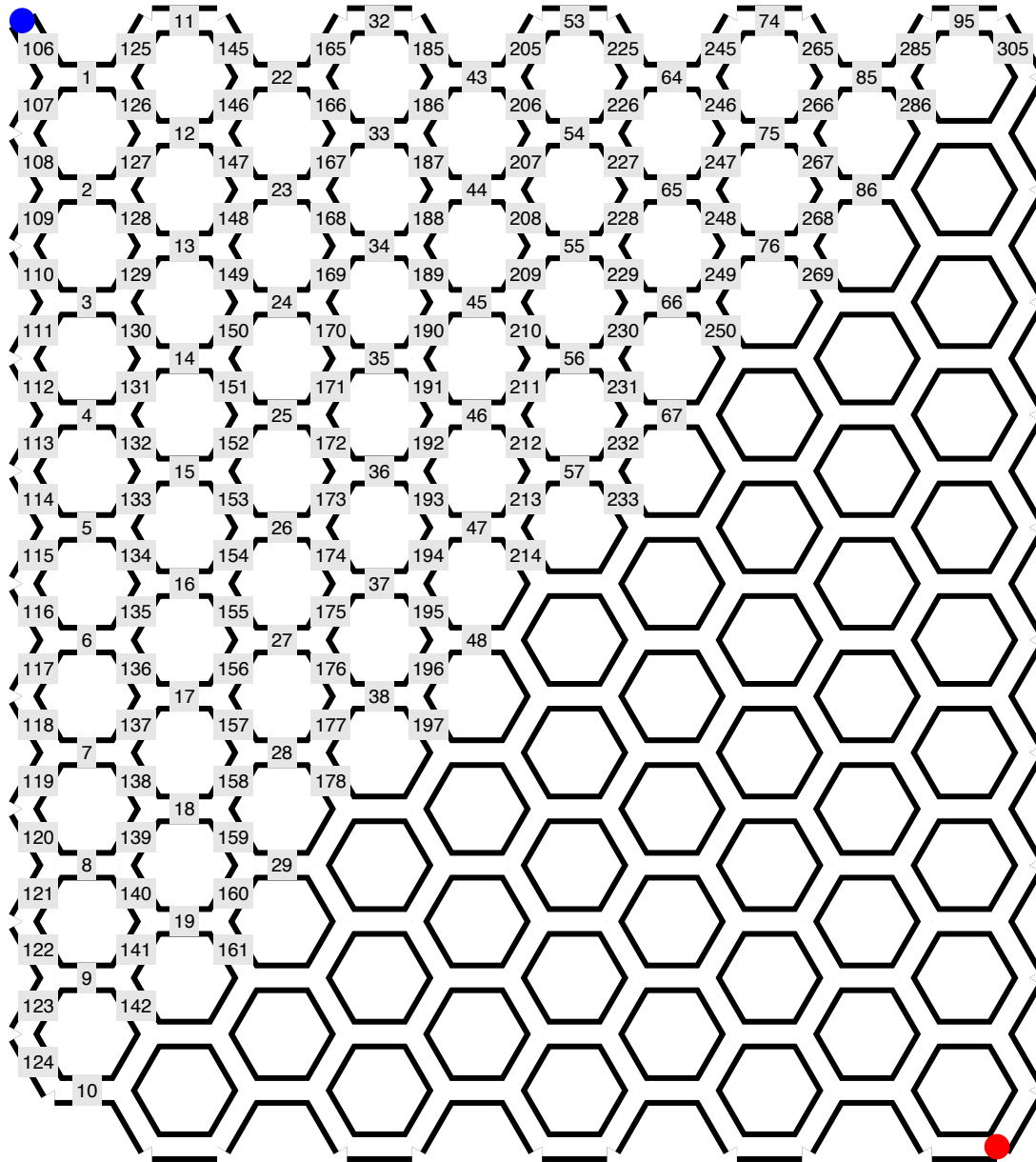


Figure 1: Capillary network with indices of vessels. Inlet is at the top left (blue dot), outlet is at the bottom right (red dot).

References

- [1] R.T. Yen and Y.C. Fung, 1978. *Effect of velocity of distribution on red cell distribution in capillary blood vessels*. Am J Physiol (**235**), H251-H257.
- [2] A.R. Pries and T.W. Secomb, 2005. *Microvascular blood viscosity in vivo and the endothelial surface layer*. Am J Physiol Heart Circ Physiol (**289**), H2657-H2664.